Evaluation report

What the synthetic test checked — and what it did not

Run date 27 October 2026 · Fixture: FICTIONAL SYNTHETIC GROUND TRUTH — not observed real-world performance

Short answer

The rules matched the labels in the one fictional hospital exercise they were written for: 8 of 8 event rows, 5 of 5 expected findings, and 9 of 9 expected evidence-record IDs. This shows that the demo pipeline works on its sample. It does not show how well the product works on a new exercise or whether a professional evaluator agrees with each conclusion.

How to read the scores

Precision asks: of the items the system proposed, what share matched the expected labels? Recall asks: of the items in the expected labels, what share did the system find? Here, all five expected finding codes were proposed and no extra code was produced, so both scores are 1.00 on this fixture. The labels were built for this same scenario; perfect fixture scores do not mean perfect real-world accuracy.

A citation match checks that the expected record ID is linked. It does not prove that the record truly supports the sentence. That semantic question still needs a human reviewer.

Worked example: power restoration

The log says power failed at 09:02 (EL-01) and essential circuits were restored at 09:06 (EL-02). The exercise target is 60 seconds. Those records support a proposed finding that restoration took about four minutes and missed the target. They do not establish the cause. The benchmark checks that the proposal and record IDs are present; an evaluator must judge whether the interpretation is fair.

What the fixture contains

Three invented files: an eight-row event log, evaluator notes with one untimed vague statement and a 09:10/09:11 disagreement, and a pre-generated transcript with an intentionally mistaken ‘too minutes’ phrase at 0.58 confidence. The transcript is not a result of audio transcription. The written labels are in demo/ground_truth.json.

Recorded technical results

Check	Result	Plain-language meaning
Event-row extraction	8/8 · precision 1.00 · recall 1.00	All eight labeled log rows were found in this sample.
Finding detection	5/5 · precision 1.00 · recall 1.00	All five expected finding labels were produced; no extra label appeared.
Unsupported proposals	0/5	Closed-fixture code/evidence check only; no human judged whether the claim was fair.
Proposals with no source link	0/5	All five proposals included at least one record ID.
Evidence-record links	9/9	Expected IDs were linked. This is not a semantic support rating.
Unresolved / unplaced times	1 conflict · 1 untimed note	The system showed the conflict and left the untimed note out of confirmed order.
Action checklist	31/32 fields	One of four action owners is intentionally blank.
Rule request time	Median 4.61 ms · 95th percentile 9.13 ms	Twelve local test calls; no AI model was loaded.
Test-process memory	78.2 MB	Memory used by the app and test harness on one Mac; excludes model use.

Edge cases checked

A note without a time stays unplaced. The 09:10 and 09:11 reports remain a visible disagreement. ‘Generator lagged a bit’ is too vague to become a finding. The low-confidence transcript mistake stays uncertain; the response-time proposal uses the separate event log.

What was not measured

No human evaluators took part. The project did not measure report-writing time, reviewer edits, evaluator agreement, cash savings, performance on other scenarios, PDF image-reading (OCR), speech-to-text word error rate, or the quality and speed of a local language model. Word error rate means the share of spoken words that a transcript gets wrong. Optional speech-to-text and local language-model paths were not run.

Comparison of approaches

Approach	Status	What that means
Manual evaluator review	Not measured	No participant, timed baseline, or record of corrections.
Fixed local rules	Measured on one fixture	Matched its labels. This does not predict results on new exercises.
Optional local writing model	Not run	No quality or speed comparison is available.
Optional speech-to-text	Not run	No real audio, word-error rate, or resource comparison is available.

Next evaluation steps

Step	What to measure
Work with evaluators	Time evidence lookup, drafting, edits, and report completion on matched exercises.
Use unseen evidence	Have evaluators label it before analysis; measure missed findings, unsupported claims, and time agreement.
Check citations	Ask independent reviewers whether each cited record actually supports each claim.
Test speech-to-text	With consented English audio, measure transcript errors, time corrections, and computer resources.
Follow actions	Measure owner completeness, closure evidence quality, and independently verified completion.

Interpretation

The demo shows that a source-linked review flow can be built with visible uncertainty and human decision gates. It does not show that AI can outperform evaluators. A real pilot must measure both correctness and reviewer effort.